Branch Coverage Report - Hostel Management System

1. Introduction

1.1 Purpose

This report gives the branch coverage of the Hostel Management System achieved by the automated test suite, as required by Mini-Project Deliverables Part-3.

1.2 Scope

It covers the six Django apps. Branch coverage counts each possible outcome of a decision (both sides of an if, entering or skipping a loop). A partial branch is a decision where at least one outcome was never taken.

1.3 References

- Software Requirements Specification (SRS) - Hostel Management System
- Software Architecture and Design Specification (SAD) - Hostel Management System
- Software Test Plan (STP) - Hostel Management System

2. Method

Coverage was measured with coverage.py during the final test run, using the same command as the CI pipeline:

```
coverage run --branch
--source=hostels,rooms,allocations,dashboard,accounts,students manage.py test
```

Application code is the six Django apps without their test files and migrations. Migrations are generated by Django and test files exercise themselves, so this is the figure that shows how well the system is tested. The CI pipeline measures the whole source tree, including tests, and fails below 80%.

2.1 Test Environment

Item	Details
Hardware	Team laptop (16 GB RAM, Intel i5 class)
Software	Windows 11, Python 3.11.9, Django 5.2.19, SQLite 3
Tools	Django test framework and test Client, coverage.py 7.17.2, GitHub Actions

3. Summary

Measure	Branches	Covered	Partial	Coverage
Application code	128	127	1	99.2%
Whole source tree	218	216	2	99.1%

4. Branch Coverage by Module

Module	Branches	Covered	Missed	Coverage
accounts	52	51	1	98.1%
allocations	34	34	0	100.0%
dashboard	0	0	0	no branches
hostels	6	6	0	100.0%
rooms	18	18	0	100.0%
students	18	18	0	100.0%

5. Branch Coverage by File

File	Branches	Covered	Partial	Coverage
accounts/decorators.py	8	8	0	100%
accounts/forms.py	10	10	0	100%
accounts/services.py	10	9	1	90%
accounts/validators.py	18	18	0	100%
accounts/views.py	6	6	0	100%
allocations/forms.py	2	2	0	100%
allocations/models.py	2	2	0	100%
allocations/services.py	18	18	0	100%
allocations/views.py	12	12	0	100%
hostels/forms.py	2	2	0	100%
hostels/views.py	4	4	0	100%
rooms/models.py	6	6	0	100%
rooms/views.py	12	12	0	100%
students/forms.py	8	8	0	100%
students/services.py	6	6	0	100%
students/views.py	4	4	0	100%

6. Partial Branches

File	Outcome not taken	Reason
accounts/services.py	line 36 to function exit	In logout_user(), the case where the request has no session. Every real request passes through the session middleware, so this safety check is never false in tests. Not a defect.
allocations/tests.py	line 278 to line 290	In UT_ALLOC_05, the retry loop using all 50 attempts. The test always finishes in a few attempts. Test code only.

7. Conclusion

Branch coverage of the application code is 99.2% (127 of 128). The one partial branch is a defensive check that cannot be reached through the web interface.